REDUCTION OF VANISHING IN ALL DEGREES TO ONE VANISHING OVER THE MAXIMAL PRO-L EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both directions of the equivalence.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Maximal pro-l subgroup of the absolute Galois group and the corresponding extension. Same chapter, the maximal pro-l subgroup H of G with the field L of the corresponding extension. Supplies the field against which the single vanishing condition is imposed.
- Comparison of Galois with etale cohomology and the behaviour of l-torsion coefficients. Same chapter, the translation into Galois cohomology and the reduction of an arbitrary l-torsion sheaf to finite coefficient pieces. Supplies the two directions of the equivalence.

Source reference. Original lemma 0DV8: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10599–10599. Navigation pointer invoked at proof line 10671; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0DV7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10543–10543. Navigation pointer invoked at proof line 10613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let ℓ be a prime number and n an integer > 0. Let K be a field with $G = Gal(K^{sep}/K)$ and let $H \subset G$ be a maximal pro-ℓ subgroup with L/K being the corresponding field extension. Then $H^q_{étale}(\mathrm{Spec}(K), \mathcal{F}) = 0$ for $q \geq n$ and all ℓ-torsion sheaves $\mathcal{F}$ if and only if $H^n_{étale}(\mathrm{Spec}(L), \underline{\mathbf{Z}/\ell\mathbf{Z}}) = 0$.*

Proof. The forward direction is trivial, so we need only prove the reverse direction. We proceed by induction on q. The case of $q = n$ is Lemma 0DV8. Now let $\mathcal{F}$ be an ℓ-power torsion sheaf on $\mathrm{Spec}(K)$. Let $f : \mathrm{Spec}(K^{sep}) \to \mathrm{Spec}(K)$ be the inclusion of a geometric point. Then consider the exact sequence:

$$0 \to \mathcal{F} \xrightarrow{res} f_* f^{-1}\mathcal{F} \to f_* f^{-1}\mathcal{F}/\mathcal{F} \to 0$$

Note that K^{sep} may be written as the filtered colimit of finite separable extensions. Thus f is the limit of a directed system of finite étale morphisms. We may, as was seen in the proof of Lemma 0DV7, conclude that f has vanishing higher direct images. Thus, we may express the higher cohomology of $f_* f^{-1}\mathcal{F}$ as the higher cohomology on the geometric point which clearly vanishes. Hence, as everything here is still ℓ-torsion, we may use the inductive hypothesis in conjunction with the long-exact cohomology sequence to conclude the result for $q + 1$. □